



READY2GO

Smart Embedded Departure Assistant

IEEE SSCS Arduino Contest 2026

Project Report

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**Embedded
Systems
Project**

Abstract

People frequently leave their homes forgetting essential personal belongings such as keys, wallets, identification cards, or work-related items. Although numerous mobile reminder applications are available, they rely entirely on manual user interaction and cannot physically verify whether these items have been taken. This limitation motivated the development of **Ready2Go**, a smart embedded departure assistant that combines real-world object verification with an interactive checklist system to improve users' readiness and reduce the likelihood of forgetting essential belongings before leaving home.

Ready2Go is designed as a standalone embedded system built around the ESP32 microcontroller. The system automatically activates using a Passive Infrared (PIR) motion sensor when the user approaches the entrance. Through a joystick-controlled interface displayed on a 16×2 I2C LCD, the user selects a destination such as University, Work, Gym, Travel, Emergency, or a custom destination. Based on the selected destination, the system dynamically generates a personalized checklist containing both regular and critical items.

To enhance reliability, Ready2Go incorporates Hall-effect sensors that physically verify the presence of critical belongings, including keys, wallet, and student ID. If any verified item is still detected at its designated storage location after the user claims it has been taken, the system immediately activates audible alerts and repeatedly verifies the item until it is physically removed. This dual-verification mechanism significantly reduces the possibility of leaving essential belongings behind while providing a higher level of reliability than conventional reminder applications.

The system also integrates with the Blynk IoT platform, providing real-time monitoring of system status, selected destination, checklist progress, readiness score, forgotten items, and activity logs. Furthermore, users can remotely create entirely new destinations and customized checklists directly from the mobile application without modifying or reprogramming the firmware. This capability allows the system to adapt to different user routines while maintaining flexible and extensible embedded architecture.

By combining embedded sensing, physical object verification, intelligent decision-making, human-machine interaction, and IoT connectivity, Ready2Go demonstrates a practical and scalable smart embedded solution for improving daily departure routines, illustrating how embedded intelligence and connected technologies can be effectively integrated to solve real-world problems.

Keywords— Embedded Systems, Internet of Things (IoT), ESP32, Smart Assistant, Hall-Effect Sensor, Physical Object Verification, Dynamic Checklist Generation, Blynk IoT, Human–Machine Interaction.

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1. Introduction

1.1 Background

In today's fast-paced lifestyle, people frequently leave their homes while forgetting important personal belongings such as keys, wallets, identification cards, laptops, chargers, or other destination-specific items. These simple mistakes often lead to unnecessary delays, missed appointments, additional transportation costs, and increased daily stress. Students may forget their university identification cards or laptops, employees may leave important work materials behind, and travelers may forget essential travel documents.

Existing reminder solutions are primarily software-based, relying on smartphone notifications or manually created to-do lists. While these applications can remind users of predefined tasks, they cannot determine whether an item has actually been taken. As a result, users often dismiss reminders without physically verifying their belongings, reducing the effectiveness of these systems in real-world scenarios.

Recent advances in Embedded Systems and the Internet of Things (IoT) have enabled the development of intelligent devices capable of interacting directly with users and their surrounding environment. By combining sensors, microcontrollers, and wireless communication, embedded systems can provide more reliable and context-aware assistance than traditional reminder applications.

This project explores how embedded technologies can be utilized to create a practical smart assistant that performs both digital checklist management and physical object verification before users leave their homes.

1.2 Motivation

The primary motivation behind Ready2Go is to improve the reliability of everyday departure routines using embedded technology.

Unlike conventional reminder systems that rely solely on user honesty or memory, Ready2Go introduces a second verification layer through physical sensing. The project aims to demonstrate how inexpensive embedded hardware components can be integrated to create a practical, intelligent assistant that minimizes human error while maintaining a simple and intuitive user experience.

Additionally, the project was developed as part of the IEEE Solid-State Circuits Society (SSCS) Arduino Contest, with the objective of showcasing an innovative embedded systems application that addresses a real-life problem through hardware-software integration.

1.3 Problem Statement

One of the most common daily problems is forgetting essential belongings before leaving home. Although mobile reminder applications are widely available, they depend entirely on manual interaction and cannot verify whether users have collected their required items.

Consequently, users may accidentally leave behind important belongings despite confirming reminders on their smartphones. This limitation becomes more significant for frequently used critical items such as keys, wallets, and identification cards, where forgetting even a single item may interrupt an entire daily schedule.

Therefore, there is a need for an embedded solution capable of not only reminding users about required items but also physically verifying the presence or absence of those belongings before departure.

1.4 Proposed Solution

Ready2Go is a standalone smart embedded departure assistant designed around the ESP32 microcontroller.

The system automatically activates when the user approaches the entrance using a Passive Infrared (PIR) motion sensor. An interactive menu displayed on a 16×2 I2C LCD allows users to select their intended destination using a joystick. Based on the selected destination, the system dynamically generates a customized checklist containing both regular and critical items.

During the checklist process, users confirm whether each item has been packed. Once the checklist is completed, Hall-effect sensors physically verify the presence of critical belongings such as keys, wallet, and student ID. If any of these items are still detected at their storage location, the system immediately issues audible alerts through an active buzzer and continues reminding the user until the item is physically removed.

To further enhance usability, Ready2Go integrates with the Blynk IoT platform, enabling real-time monitoring of system status, checklist progress, readiness score, activity logs, and forgotten items. The mobile application also allows users to create entirely new destinations and customized checklists without modifying or reprogramming the firmware.

1.5 Objectives

The primary objectives of Ready2Go are summarized as follows:

- Design a standalone embedded assistant that operates without requiring continuous smartphone interaction.
- Automatically detect user presence before departure using motion sensing.

- Generate destination-specific checklists dynamically.
- Distinguish between critical and non-critical belongings.
- Physically verify essential belongings using Hall-effect sensors.
- Reduce the probability of forgetting important personal items.
- Calculate a readiness score based on user interaction and verification results.
- Provide real-time monitoring through the Blynk IoT platform.
- Allow users to create fully customized destinations and personalized checklists remotely without firmware modification.
- Demonstrate the integration of Embedded Systems and IoT technologies in solving practical everyday problems.

2. Problem Statement

2.1 Traditional Reminder Applications

Reminder applications are among the most common solutions used to reduce the likelihood of forgetting daily tasks or personal belongings. Applications such as Google Keep, Microsoft To-Do, Apple Reminders, and Todoist allow users to create checklists, schedule notifications, and receive location- or time-based reminders.

These applications offer several advantages, including ease of use, cloud synchronization, and cross-device accessibility. Users can organize daily routines and receive notifications before leaving a location.

However, these systems rely entirely on manual user interaction. They cannot determine whether an item has been packed or physically carried by the user. Once a reminder is dismissed, the application assumes the task has been completed, even if the required object remains behind.

As a result, reminder applications function only as digital memory aids rather than verification systems.

2.2 Smart Home Reminder Systems

Recent developments in smart home technology have introduced intelligent assistants capable of automating household tasks and providing voice-based reminders. Platforms such as **Amazon Alexa**, **Google Home**, and **Apple HomeKit** can announce reminders, respond to voice commands, and interact with other smart devices.

Although these systems improve user convenience, they primarily depend on cloud services and voice interaction. They lack direct physical awareness of users' belongings and cannot verify whether important objects have actually been taken before leaving home.

Furthermore, most smart home ecosystems require expensive infrastructure and continuous internet connectivity, making them less practical for simple embedded reminder applications.

2.3 RFID-Based Reminder Systems

RFID technology has been widely used in research projects designed to reduce forgotten belongings. In these systems, important objects are equipped with RFID tags, while an RFID reader scans for the presence of tagged items before the user leaves.

Although RFID provides automatic object identification, it introduces several limitations. Every object must carry an RFID tag, increasing system cost and maintenance. Tags may become damaged or detached over time, and RFID readers often require careful placement and calibration to achieve reliable detection.

Additionally, RFID systems focus on tag detection rather than natural user interaction, making them less intuitive for everyday household use.

2.4 Existing Commercial Solutions

Several commercial products attempt to solve the problem of forgotten belongings through Bluetooth tracking devices such as **Apple AirTag**, **Samsung SmartTag**, and **Tile**.

These products excel at locating lost items after they have already been misplaced. However, they are not designed to prevent users from forgetting belongings before leaving home.

Similarly, Bluetooth-based reminder applications notify users only after the distance between the user and the tracked object exceeds a predefined threshold. At that point, the user has already left the house, reducing the effectiveness of the reminder.

Therefore, these solutions emphasize item recovery rather than departure verification.

2.5 Comparative Analysis

Feature	Reminder Apps	Smart Home Assistants	RFID Systems	Bluetooth Trackers	Ready2Go
Interactive Checklist	✓	Limited	✗	✗	✓
Destination-Based Checklist	✗	✗	✗	✗	✓
Physical Object Verification	✗	✗	Partial	Partial	✓
Critical Item Detection	✗	✗	Limited	✗	✓
Real-Time Readiness Score	✗	✗	✗	✗	✓
IoT Dashboard	Limited	✓	Limited	Mobile	✓
Custom Destinations	✗	✗	✗	✗	✓
Embedded Standalone Device	✗	✗	✓	✗	✓
Immediate Audible Alerts	✗	Voice Only	Limited	After Leaving	✓
Low Hardware Cost	✓	✗	Medium	Medium	✓

2.6 Why Ready2Go?

Ready2Go was designed to overcome the limitations observed in existing reminder technologies by combining the strengths of embedded systems, physical sensing, and IoT connectivity into a single integrated solution.

Unlike conventional reminder applications that rely solely on user confirmation, Ready2Go performs **physical verification** of essential belongings using Hall-effect sensors. This additional verification layer significantly improves reliability by ensuring that critical objects are actually removed from their designated storage locations before departure.

The system also introduces **destination-aware checklists**, allowing different item lists to be automatically generated for locations such as university, work, gym, travel, emergency situations, or fully customized destinations created directly through the Blynk mobile application without modifying the firmware.

Another unique contribution of Ready2Go is its **hybrid embedded-IoT architecture**. The embedded prototype operates independently without requiring continuous cloud connectivity, while the Blynk dashboard provides real-time monitoring, checklist progress, readiness score visualization, forgotten-item reporting, and activity logging.

Furthermore, Ready2Go focuses on **preventing mistakes before they occur**, rather than helping users recover forgotten belongings afterward. By combining intelligent interaction, hardware verification, and customizable workflows, the proposed system offers a practical and user-friendly solution that is better suited for everyday departure routines than existing alternatives.

2.7 Chapter Summary

This chapter reviewed several categories of existing reminder technologies, including reminder applications, smart home assistants, RFID-based systems, and commercial Bluetooth tracking devices. While each solution addresses part of the forgetting problem, none provides a complete combination of destination-aware checklists, physical verification of belongings, embedded operation, and IoT-based monitoring.

Ready2Go integrates these capabilities into a unified embedded system, offering a more reliable and practical approach for preventing forgotten belongings before users leave their homes.

3. System Requirements

3.1 Functional Requirements

Ready2Go system was designed to satisfy the following functional requirements:

FR1 – User Presence Detection

The system shall automatically detect the presence of a user using a PIR motion sensor and wake up from sleep mode.

FR2 – Destination Selection

The system shall allow the user to select a destination through the joystick interface.

Supported destinations include:

University, Work, Gym, Travel, Emergency, and Custom Destination

FR3 – Dynamic Checklist Generation

The system shall generate a checklist based on the selected destination.

Each destination shall have its own predefined list of required items.

FR4 – Critical Item Management

The system shall distinguish between:

Non-Critical Items

Critical Items (*Critical items cannot be skipped without generating warnings and reducing the readiness score*).

FR5 – Physical Item Verification

The system shall verify that important belongings are physically taken using Hall-effect sensors.

Items verified include:

Keys, Wallet, Student ID

If an item is still detected on its storage location, the system shall continuously alert the user until it is removed.

FR6 – Readiness Score Calculation

The system shall calculate a readiness score between 0 and 100%, and the score decreases according to:

- Skipped critical items
- Skipped optional items
- Physically forgotten belongings

FR7 – User Feedback

The system shall provide immediate feedback through:

- LCD Display
- Active Buzzer
- Blynk Mobile Dashboard

FR8 – Session Logging

The system shall record session information including:

Destination, Readiness Score, Missing Items, and Session Duration

FR9 – IoT Monitoring

The system shall transmit real-time information to the Blynk application, including:

- Current System Status
 - Selected Destination
 - Current Checklist Item
 - Missing Items
 - Readiness Score
 - Assistant Messages
-

FR10 – Custom Destination Feature

The system shall allow users to create new destinations directly from the mobile application without modifying the firmware.

Users shall also be able to define custom checklist items and specify which items are critical.

3.2 Non-Functional Requirements

Besides the functional behavior, Ready2Go was designed to satisfy several non-functional requirements.

Performance

- Fast system response after motion detection.
 - Immediate user feedback during interaction.
 - Low latency between ESP32 and Blynk dashboard.
-

Reliability

- Physical verification reduces false confirmations.
 - Hall-effect sensing ensures accurate detection of critical belongings.
 - Automatic validation prevents accidental departures.
-

Usability

- Simple joystick navigation.
 - Clear LCD interface.
 - Minimal learning curve.
 - Mobile dashboard for remote monitoring.
-

Scalability

Software architecture allows future expansion, including:

- Additional destinations
 - More sensors
 - Cloud databases
 - AI recommendation systems
 - Voice assistant integration
-

Maintainability

The software was written using modular programming principles.

Separate helper functions were implemented for:

- LCD management
- Hardware verification
- Blynk communication
- Score calculation
- State management

This simplifies debugging and future updates.

Energy Efficiency

The system remains in sleep mode until motion is detected by the PIR sensor, reducing unnecessary power consumption.

Cost Effectiveness

The project relies on inexpensive and widely available components, making it suitable for practical deployment and educational purposes.

Security

Only authenticated Blynk users can modify custom destinations and personalized checklists.

4. System Design

4.1 Overall System Architecture

The Ready2Go system adopts a modular embedded architecture centered around the ESP32 microcontroller, which coordinates user interaction, decision making, hardware verification, and IoT communication. The operation starts when a PIR sensor detects the user's presence, activating the system and allowing destination selection through the joystick. Based on the selected or custom destination, the system generates a dynamic checklist and classifies items according to their priority. Critical belongings are then physically verified using Hall-effect sensors before calculating the final readiness score. Throughout the entire process, system status, checklist progress, missing items, and session information are synchronized with the Blynk IoT dashboard, providing real-time monitoring and remote customization capabilities.

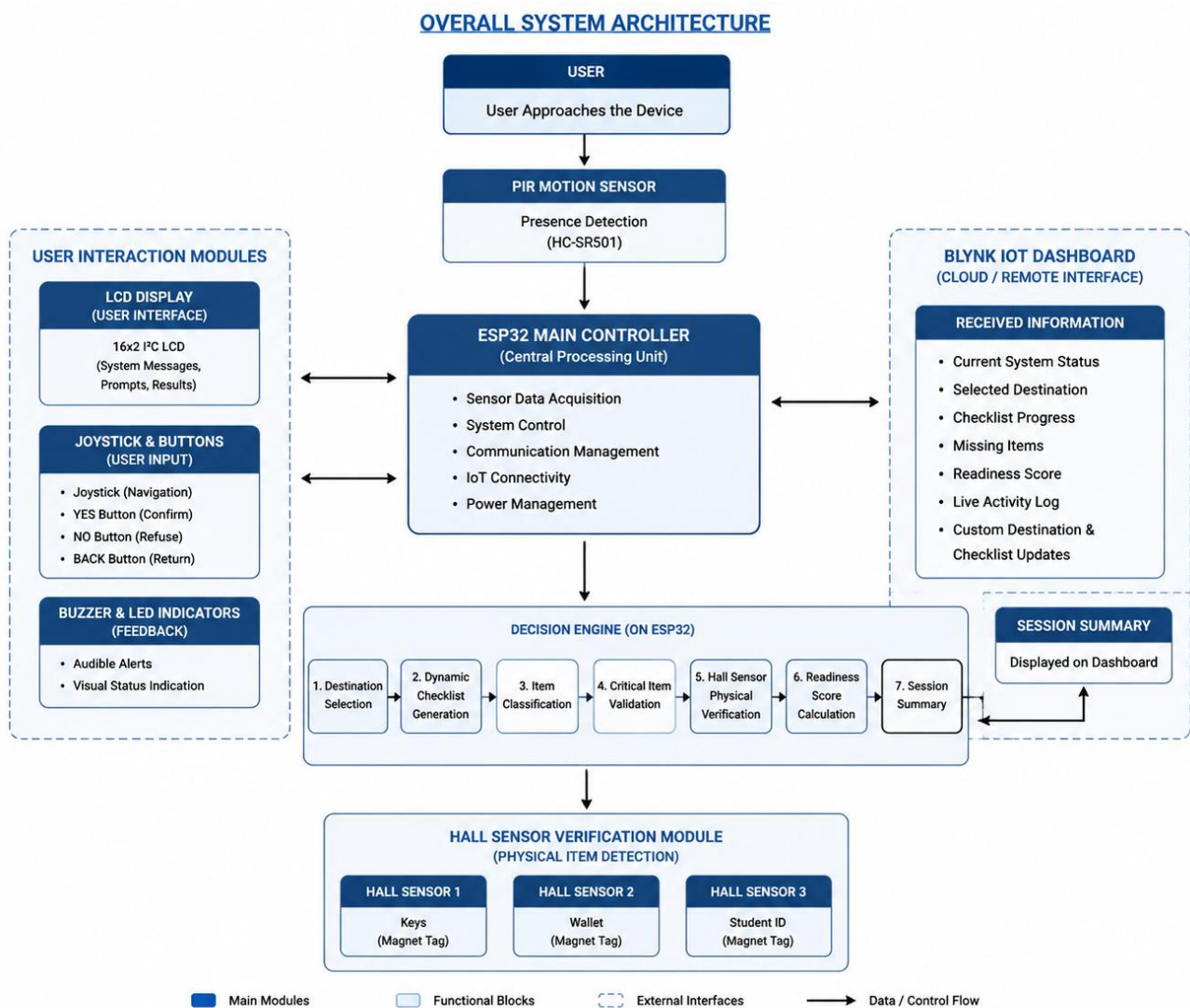


Figure 1. Overall system architecture of Ready2Go – Smart Embedded Departure Assistant.

4.2 Hardware Design

4.2.1 Hardware Overview

The Ready2Go prototype was implemented as a compact embedded system built around the ESP32 microcontroller. The hardware integrates multiple sensing, interaction, and feedback components to assist users before leaving home. A PIR motion sensor automatically activates the system when user presence is detected, while a joystick and push buttons provide an intuitive interface for destination selection and checklist navigation. An LCD module displays system prompts and feedback throughout the session. Hall-effect sensors perform physical verification of critical belongings such as keys, wallet, and student ID, ensuring that these items have been taken before departure. An active buzzer provides audible alerts whenever important items are forgotten. In addition, the ESP32's built-in Wi-Fi capability enables seamless communication with the Blynk IoT platform for real-time monitoring and custom destination management.



4.2.2 Bill of Materials (BOM)

Component	Qty	Purpose	Approx. Cost (L.E)
ESP32 Devkit V1	1	Main microcontroller	350
USB Cable	1	Programming and Power Supply	80
LCD 2×16 I2C	1	User interface display	145
Joystick Module	1	Destination selection and menu navigation	50
Push Button (Green)	1	Confirm / Yes	10
Push Button (Red)	1	Rejection / No	10
Push Button	1	Back / Return	10
PIR Motion Sensor (HC-SR501)	1	User presence detection	50
Active Buzzer	1	Audible alerts	10
Breadboard	1	Prototype assembly	35
Jumper Wires	1 Set	Electrical connections	55
Hall Effect Sensor (A3144)	3	Physical verification of critical belongings	45
Neodymium Magnets	3	Trigger Hall-effect sensors	20
TOTAL PROTOTYPE COST		1000 EGP	

4.2.3 Component Selection

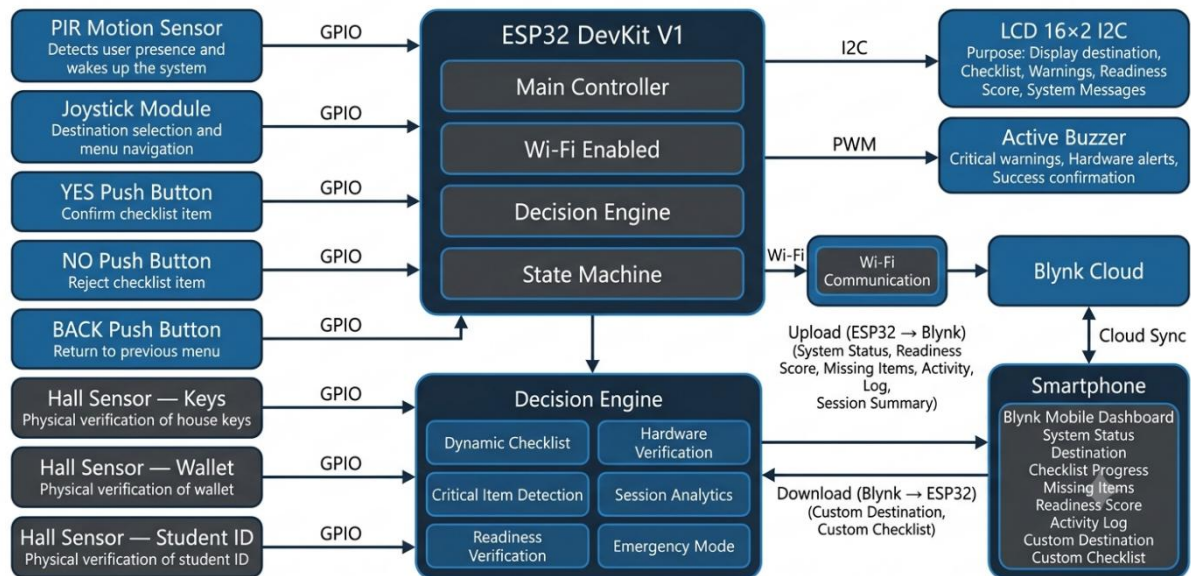
The hardware components used in the Ready2Go prototype were selected to achieve a balance between functionality, cost, reliability, and ease of integration. Since the project targets a practical low-cost embedded solution, commonly available components with stable software support and simple interfacing were preferred. Table 4.2 summarizes the rationale behind selecting each major hardware component.

Component	Reason for Selection
ESP32 DevKit V1	Selected as the main controller because it provides sufficient processing power, built-in Wi-Fi for IoT connectivity, multiple GPIO pins, and excellent compatibility with the Arduino ecosystem.
LCD 16×2 I2C	Chosen to provide a simple and readable user interface while minimizing GPIO usage through the I2C communication protocol.
PIR Motion Sensor (HC-SR501)	Used to automatically detect user presence, allowing the system to remain idle until someone approaches, reducing unnecessary operation.
Joystick Module	Selected as an intuitive navigation device for browsing destinations and interacting with menus without requiring a touchscreen.
Push Buttons	Provide dedicated actions (Yes, no, and back), simplifying user interaction and improving usability.
Hall Effect Sensors (A3144)	Chosen for reliable, contactless detection of critical belongings such as keys, wallet, and student ID using magnetic sensing.
Neodymium Magnets	Attached to important personal belongings to enable accurate Hall-effect sensor detection.
Active Buzzer	Provides immediate audible alerts whenever critical items are forgotten, ensuring the user notices warnings even without looking at the display.
Breadboard & Jumper Wires	Used for rapid prototyping, testing, and easy hardware modifications during development.

4.2.4 Hardware Block Diagram

The Ready2Go hardware architecture is centered around the ESP32 Devkit V1, which acts as the main processing and communication unit. All input devices, including the PIR motion sensor, joystick, push buttons, and Hall-effect sensors, are directly connected to the ESP32 through its GPIO pins. Based on the received sensor data and user interactions, the ESP32 executes the decision-making algorithm, updates the LCD display, activates the buzzer when necessary, and synchronizes system information with the Blynk IoT platform via Wi-Fi. This centralized architecture simplifies system integration while maintaining modularity, allowing additional sensors or communication modules to be incorporated with minimal hardware modifications.

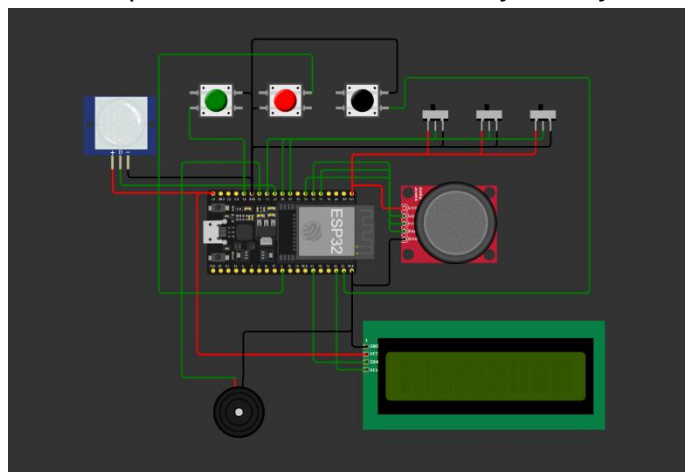
Hardware Block Diagram of the Ready2Go Smart Embedded Departure Assistant



4.2.5 Circuit Diagram (Wokwi)

To validate the proposed hardware architecture before physical implementation, the complete system was first designed and tested using the Wokwi online simulation platform. The simulation environment provided a safe and efficient method for verifying hardware connections, firmware functionality, and system behavior prior to assembling the physical prototype.

The Wokwi model includes all major hardware components used in the Ready2Go system, including the ESP32 Devkit V1, LCD 2x16 I2C display, PIR motion sensor, joystick module, push buttons, Hall-effect sensors, active buzzer, and the required interconnections. This simulation was extensively used during firmware development to verify the finite-state machine, checklist workflow, destination selection, hardware verification logic, and user interface before deployment on the actual hardware.



After successful simulation, the same wiring configuration was transferred directly to the physical prototype with only minor adjustments required for hardware assembly.

Figure 4.x illustrates the complete Wokwi circuit used during the development process.

4.2.6 ESP32 Pin Mapping

The ESP32 Devkit serves as the central controller of the Ready2Go system. Each peripheral is assigned to a dedicated GPIO pin according to its electrical characteristics and functional requirements. Digital input pins are used for push buttons and Hall-effect sensors, while an analog input pin is allocated to the joystick for menu navigation. The I²C bus is utilized for communication with the LCD display to minimize GPIO consumption. Table 4.2 summarizes the complete pin allocation used in the prototype.

ESP32 GPIO	Connected Component	Direction	Function
GPIO 21	LCD SDA	I/O	I ² C Data
GPIO 22	LCD SCL	Output	I ² C Clock
GPIO 27	PIR Motion Sensor	Input	Detects user presence
GPIO 34	Joystick X-axis	Analog Input	Destination navigation
GPIO 32	Joystick Push Button	Input	Destination selection
GPIO 13	YES Push Button	Input	Confirm checklist item
GPIO 5	NO Push Button	Input	Reject checklist item
GPIO 23	BACK Push Button	Input	Return to previous menu
GPIO 25	Hall Sensor (Keys)	Input	Physical verification of keys
GPIO 26	Hall Sensor (Wallet)	Input	Physical verification of wallet
GPIO 14	Hall Sensor (Student ID)	Input	Physical verification of student ID
GPIO 12	Active Buzzer	Output	Audio alerts and notifications
3.3 V	Sensors & LCD	Power	Supplies peripheral modules
GND	All Components	Ground	Common reference

The selected GPIO assignments were chosen to avoid hardware conflicts while maintaining compatibility with the ESP32 architecture. Analog-capable GPIO34 is dedicated to joystick position sensing, whereas the remaining peripherals utilize digital GPIOs configured with internal pull-up resistors where appropriate. Communication with the LCD is performed over the I²C interface, reducing wiring complexity and preserving GPIO resources for future expansion. This pin allocation also facilitates debugging, improves hardware organization, and allows additional sensors or communication modules to be integrated without significant modifications to the existing design.

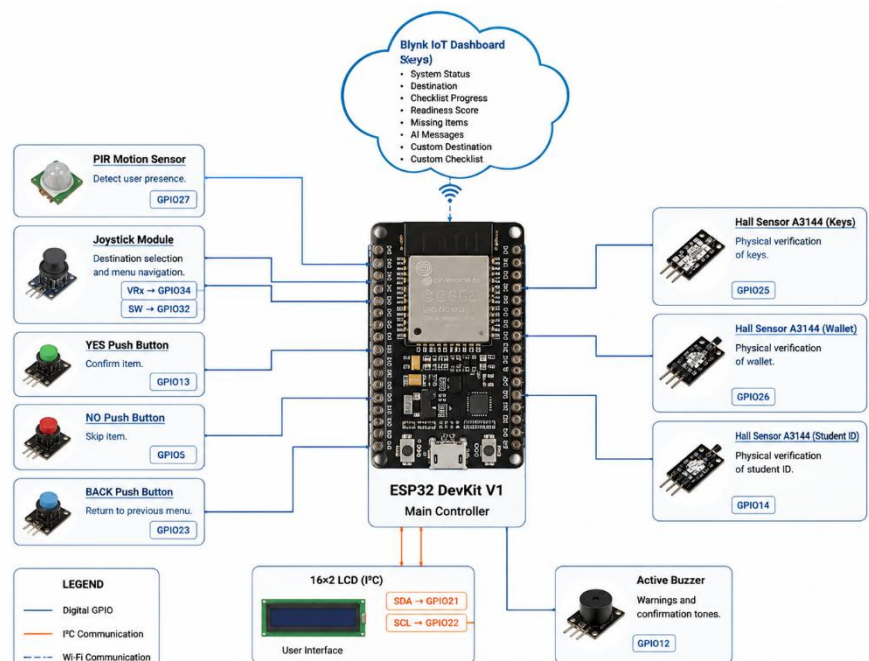


Figure 4.X ESP32 Pin Mapping of the Ready2Go Embedded System

4.3 Software Design

The software architecture of the Ready2Go system was designed using a modular approach to simplify development, improve code maintainability, and facilitate future expansion. The firmware was developed using the Arduino framework for the ESP32 platform and follows a layered architecture in which each software module is responsible for a specific task.

Instead of implementing the application as a sequence of independent functions, the system is organized around a finite-state machine that controls the complete user workflow from motion detection to session completion. Additional software modules manage destination selection, checklist generation, hardware verification, IoT communication, readiness score calculation, and user interaction through the LCD and input devices.

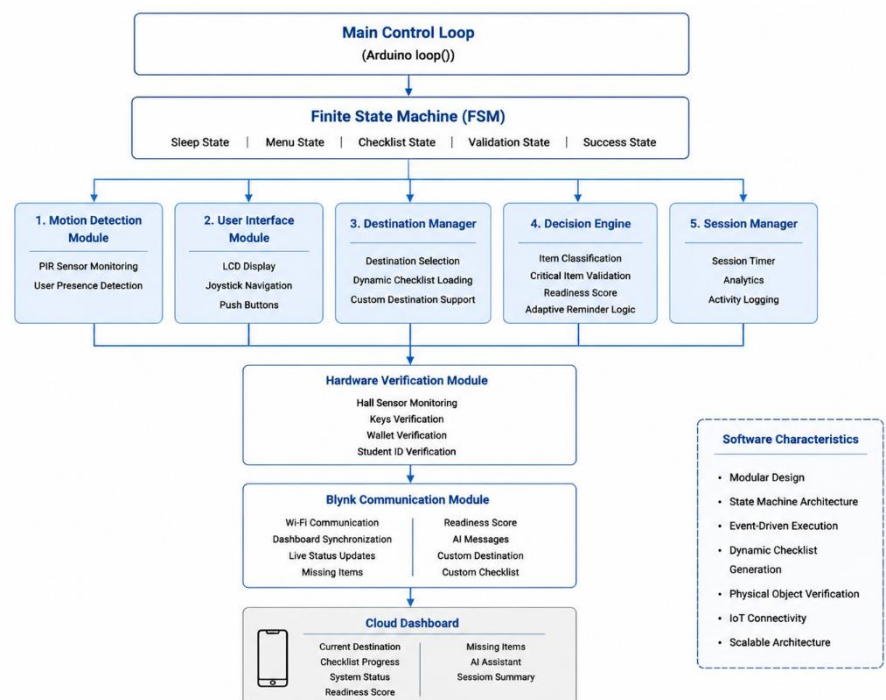
This modular architecture reduces software complexity, improves debugging, and enables new features to be integrated with minimal modification to the existing codebase.

4.3.1 Software Architecture

The Ready2Go firmware consists of several independent software modules that cooperate to execute the complete system workflow. Each module performs a dedicated responsibility while exchanging data through shared system variables and state transitions.

The software architecture is composed of the following modules:

- Motion Detection Module
- User Interface Module
- Destination Manager
- Checklist Generator
- Decision Engine
- Hardware Verification Module
- Readiness Score Calculator
- Blynk Communication Module
- Session Manager



This modular organization simplifies future maintenance and allows individual software components to be modified without affecting the remaining system.

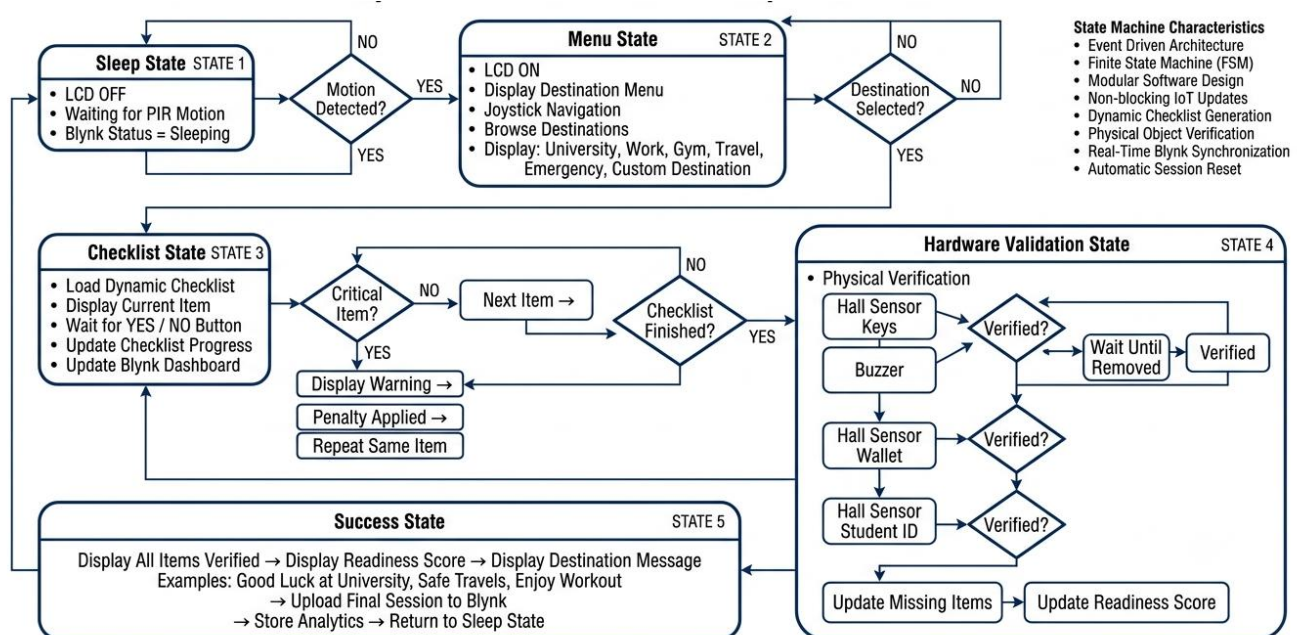
4.3.2 State Machine

The Ready2Go firmware is implemented as a finite-state machine (FSM), where each operating mode is represented as an independent software state. The state machine guarantees deterministic system behavior and eliminates unnecessary software complexity compared to traditional sequential programming.

The system begins in the Sleep State, where it continuously monitors the PIR motion sensor while minimizing processing activity. After detecting user presence, the firmware transitions to the Menu state, allowing the user to select the desired destination.

Following destination selection, the Checklist state dynamically generates the required checklist according to the selected destination. Once all checklist items are processed, the Hardware Verification state validates the presence of critical belongings using Hall-effect sensors.

Finally, the Success state displays the readiness score, session summary, and farewell message before returning to the Sleep state to await the next user.



4.3.3 Data Structures

The Ready2Go firmware was designed using lightweight and efficient data structures suitable for embedded systems with limited memory resources. Instead of relying on dynamic memory allocation or complex object-oriented containers, the system utilizes static arrays, enumerations, pointers, and structured variables to achieve predictable memory usage and deterministic execution time.

This design choice improves execution speed, minimizes memory fragmentation, and ensures stable operation on the ESP32 microcontroller throughout long-term execution.

The following data structures constitute the core software architecture of the system.

A. Destination Database

Each predefined destination is stored inside a string array.

The selected destination is accessed using an index variable (menuIndex), allowing fast navigation through the joystick interface without requiring conditional branching for every option.

```
String destinations[] =
{
    "University",
    "Work",
    "Gym",
    "Travel",
    "Emergency",
    "Custom"
};
```

B. Checklist Arrays

Each destination has two parallel arrays:

- Item names
- Criticality flags

Example:

```
String uniItems[]
{
    "Laptop",
    "Charger",
    "Case",
    "Notebook",
    "Keys",
    "Wallet",
    "Student ID"
};

bool uniCritical[]
{
    false,
    false,
    false,
    false,
    true,
    true,
    true
};
```

The parallel-array structure enables direct mapping between every checklist item and its importance level while keeping memory consumption minimal.

C. Dynamic Checklist Pointers

Instead of copying arrays into memory, Ready2Go switches between checklists using pointers.

Depending on the selected destination, these pointers reference the corresponding checklist arrays.

```
String* currentItems;

bool* currentCritical;
```


Example:

```
currentItems = uniItems;  
  
currentCritical = uniCritical;
```

This pointer-based architecture eliminates unnecessary memory duplication while allowing all destinations to share the same checklist-processing algorithm.

D. System State Enumeration

The firmware follows a finite state machine implemented using enumeration.

Using enumerations improves code readability, simplifies state transitions, and avoids the use of arbitrary numeric constants.

```
enum SystemState  
{  
    STATE_SLEEP,  
    STATE_MENU,  
    STATE_CHECKLIST,  
    STATE_VALIDATE,  
    STATE_SUCCESS  
};
```

E. Custom Destination Storage

To support user-defined destinations received from the Blynk mobile application, the system allocates dedicated arrays.

These arrays are updated dynamically after receiving data from the cloud, enabling users to create entirely new destinations without modifying or reprogramming the firmware.

```
String customChecklist[15];  
  
bool customCritical[15];  
  
int customItemsCount;
```

F. Runtime Variables

Several global variables store the current session status.

Examples include:

- Current destination
- Current checklist index
- Readiness score
- Missing items
- Session duration
- Assistant messages

These variables allow continuous synchronization between the embedded firmware, LCD interface, and Blynk dashboard during system operation.

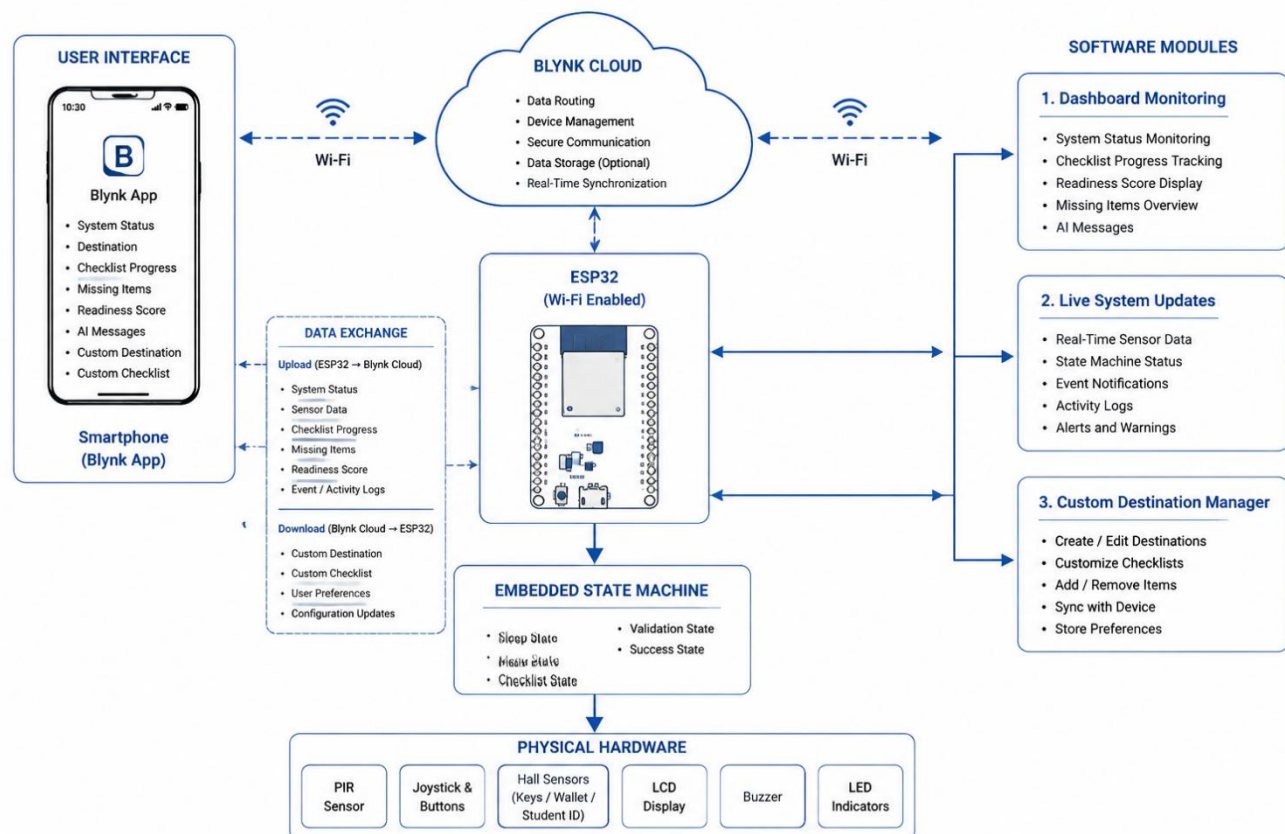
4.3.4 Blynk Communication Module

The Ready2Go system employs the Blynk IoT platform to establish wireless communication between the ESP32 and the mobile application through the built-in Wi-Fi interface. The communication module synchronizes important system events using Blynk Virtual Pins, enabling real-time data exchange without affecting the embedded system's operation.

Besides monitoring system status, the communication module enables one of the project's key features: **Custom Destination**. Users can remotely create new destinations, define personalized checklists, and specify critical items directly from the Blynk application. These parameters are transmitted to the ESP32, parsed dynamically, and immediately integrated into the destination menu without modifying or re-uploading the firmware.

Virtual Pin	Function
V0	System Status
V1	Current Destination
V2	Current Checklist Item
V3	Readiness Score
V4	AI Assistant Messages
V5	Missing Items
V6	Custom Destination Name
V7	Custom Checklist & Critical Items

This lightweight communication architecture extends the functionality of Ready2Go from a standalone embedded device to a connected IoT assistant while maintaining low computational overhead and real-time performance.



4.4 Decision Engine

The Decision Engine is the core software component responsible for coordinating the system workflow after the user interacts with Ready2Go. Instead of following a fixed sequence, the engine evaluates the selected destination, generates the appropriate checklist, classifies each item according to its importance, verifies essential belongings using hardware sensors, and finally calculates the user's readiness score before allowing the session to finish.

The decision-making process combines software logic with physical sensing, enabling the system to react differently according to user responses and real-world conditions rather than relying solely on manual confirmations.

4.4.1 Destination Selection

The first decision made by the system is identifying the user's intended destination. The destination is selected through the joystick interface, while custom destinations can be transmitted remotely using the Blynk mobile application.

Each destination corresponds to a predefined checklist stored in the firmware. Once a destination is confirmed, the Decision Engine automatically loads the associated checklist without requiring any manual configuration.

For custom destinations, the ESP32 dynamically parses the checklist received from the Blynk cloud, allowing users to create new travel scenarios without modifying or reprogramming the firmware.

4.4.2 Dynamic Checklist Generation

After the destination is selected, the Decision Engine automatically generates a destination-specific checklist. Each predefined destination is associated with a dedicated list of required items stored inside the firmware, allowing the system to adapt its behavior according to the user's activity.

Unlike conventional reminder applications that rely on static lists, Ready2Go supports dynamic checklist generation through the Blynk IoT platform. Users can create completely new destinations, define custom items, and specify which items should be treated as critical directly from the mobile application. The ESP32 receives the transmitted data, passes the checklist, and immediately integrates it into the decision process without requiring firmware modifications.

This mechanism makes the system easily extensible while maintaining a simple embedded implementation.

Destination	Generated Checklist (Example)
University	Laptop, Notebook, Keys, Wallet, Student ID
Gym	Water Bottle, Keys, Wallet
Travel	Passport, Travel Bag, Wallet
Custom (Blynk)	User-defined items

4.4.3 Item Classification

Each checklist item is assigned a priority level before the verification process begins. The Decision Engine classifies items into two categories:

- **Critical Items:** Essential belongings that should never be forgotten, such as keys, wallet, and student ID.
- **Non-Critical Items:** Optional belongings whose absence does not prevent the user from completing the session.

This classification directly influences the system's behavior. If a non-critical item is skipped, the system simply deducts a small number of readiness points and proceeds to the next item. However, when a critical item is skipped, the system immediately generates an audible warning, displays a notification on the LCD, reduces the readiness score, and repeatedly asks the user to confirm the item before continuing.

This priority-based decision mechanism improves reliability by focusing user attention on the most important belongings.

Item	Classification
Keys	Critical
Wallet	Critical
Student ID	Critical
Laptop	Non-Critical

4.4.4 Hardware Verification

After the checklist is completed, Ready2Go performs a second verification stage using Hall-effect sensors. Unlike conventional reminder systems that rely only on user input, this stage validates whether essential belongings have been physically removed from their designated locations.

Each monitored object contains a small neodymium magnet, while an A3144 Hall-effect sensor is installed at its storage location. If the sensor still detects the magnet after the user claims to have taken the item, the Decision Engine identifies the inconsistency, activates the buzzer, displays a warning message on the LCD, updates the Blynk dashboard, and waits until the object is physically removed.

This hardware-assisted verification significantly improves system reliability by preventing accidental confirmation of forgotten belongings.

4.4.5 Decision Logic

The Decision Engine is responsible for coordinating all system decisions throughout the departure process. After receiving the selected destination, the engine dynamically loads the corresponding checklist and processes user responses to one item at a time. Each checklist

item is classified as either critical or non-critical, allowing the system to apply different actions depending on its importance.

If a critical item is skipped, the system immediately displays a warning message, activates the buzzer, and prevents the user from proceeding until the item is acknowledged. Non-critical items can be skipped, but doing so reduces the overall readiness score.

Once the checklist is completed, the Decision Engine initiates hardware verification using Hall-effect sensors. If any essential belonging is still detected on its designated location, the system repeatedly alerts the user until the item is physically removed. Only after all verification steps have been completed does the system calculate the final readiness score and terminate the session successfully.

This rule-based decision process ensures that Ready2Go combines user interaction with physical object verification to minimize the possibility of forgetting essential belongings.

4.4.6 Readiness Score Calculation

The Readiness Score provides a simple numerical indication of how well the user completed the departure preparation process. Each session starts with a score of **100%**, which is gradually reduced according to user actions and hardware verification results.

The scoring algorithm applies different penalties depending on the severity of each event. Skipping a critical checklist item results in a larger score reduction than skipping a non-critical item. Likewise, leaving an essential object such as the keys, wallet, or student ID on its designated holder incurs the highest penalty because the system physically verifies these items using Hall-effect sensors.

The final score is clamped between **0% and 100%** and displayed on both the LCD and the Blynk dashboard at the end of every session. Besides providing immediate feedback to the user, the readiness score can also be used for future analytics and long-term habit monitoring.

The scoring policy implemented in Ready2Go is summarized below.

Event	Penalty
Initial Score	100
Skip Critical Item	-15
Skip Non-Critical Item	-5
Essential Item Still Detected by Hall Sensor	-20
Minimum Score	0

4.4.7 User Interface Design

The Ready2Go user interface was designed with simplicity and ease of interaction in mind. Since the system is intended to operate near the entrance of a home, the interface

minimizes user input while providing clear visual feedback throughout the entire departure process.

The LCD serves as the primary interaction interface. After motion is detected, the display presents the destination selection menu, allowing the user to navigate using the joystick. Once a destination is selected, the LCD sequentially displays each checklist item and waits for user confirmation through the YES and NO buttons.

During hardware verification, the interface provides real-time feedback indicating which essential item is still detected on its designated holder. After successful verification, the system displays the final readiness score followed by a destination-specific farewell message.

In parallel with the embedded interface, the Blynk mobile dashboard provides remote monitoring capabilities by displaying the current destination, checklist progress, system status, readiness score, missing items, and session log. Users can also create completely new destinations and personalized checklists directly from the mobile application without modifying the firmware, making the interface both flexible and user-friendly.

4.5 User Interface Design

4.5.1 LCD Interface

The 2×16 I2C LCD serves as the primary user interface of the system. It provides clear visual feedback during every stage of operation without requiring a smartphone.

The LCD displays:

- Welcome and system initialization messages.
- Destination selection menu.
- Checklist items with progress indication.
- Hardware verification status.
- Readiness score.
- Final departure confirmation message.



The interface was intentionally designed using short, readable messages to fit within the display limitations while maintaining a smooth user experience.

4.5.2 User Input (Joystick & Buttons)

Ready2Go uses a joystick and three push buttons to provide an intuitive navigation system without requiring a touchscreen or mobile device.

The joystick is used to browse available destinations, while the buttons perform dedicated actions:

Component	Function
Joystick	Navigate destination menu
YES Button	Confirm checklist item
NO Button	Skip non-critical item
BACK Button	Return to previous menu

Critical items cannot be skipped without generating warnings and reducing the readiness score, ensuring that essential belongings receive additional attention.

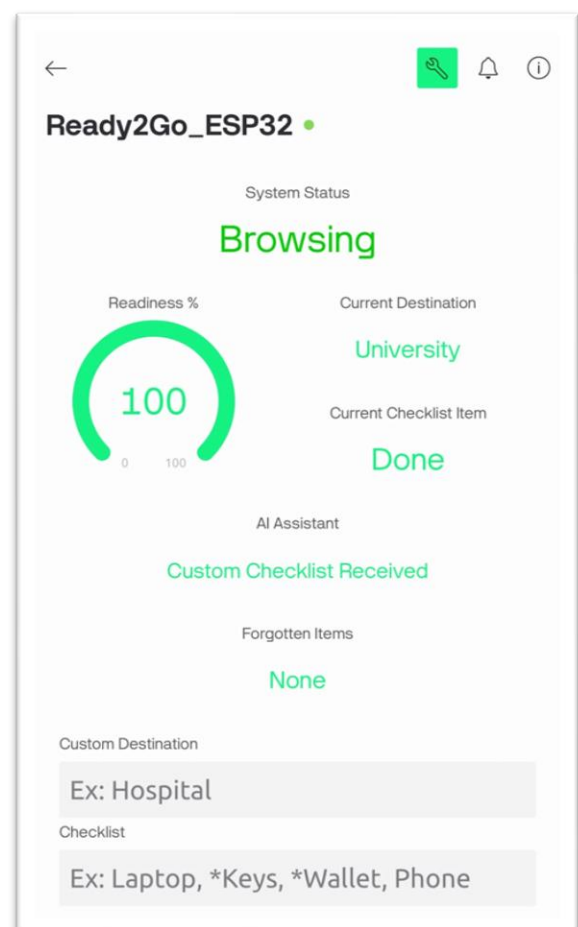
4.5.3 Blynk IOT Dashboard

To extend the capabilities of the embedded system, Ready2Go integrates with the Blynk IoT platform, enabling real-time monitoring and remote customization.

The dashboard displays:

- Current system status.
- Selected destination.
- Current checklist item.
- Readiness score.
- AI assistant messages.
- Missing items detected by Hall sensors.

In addition, users can create entirely new destinations directly from the Blynk application by specifying a destination name and a customized checklist. Critical items can also be marked from the application without modifying the firmware or reprogramming the ESP32, making the system highly flexible and easily adaptable to different user scenarios.

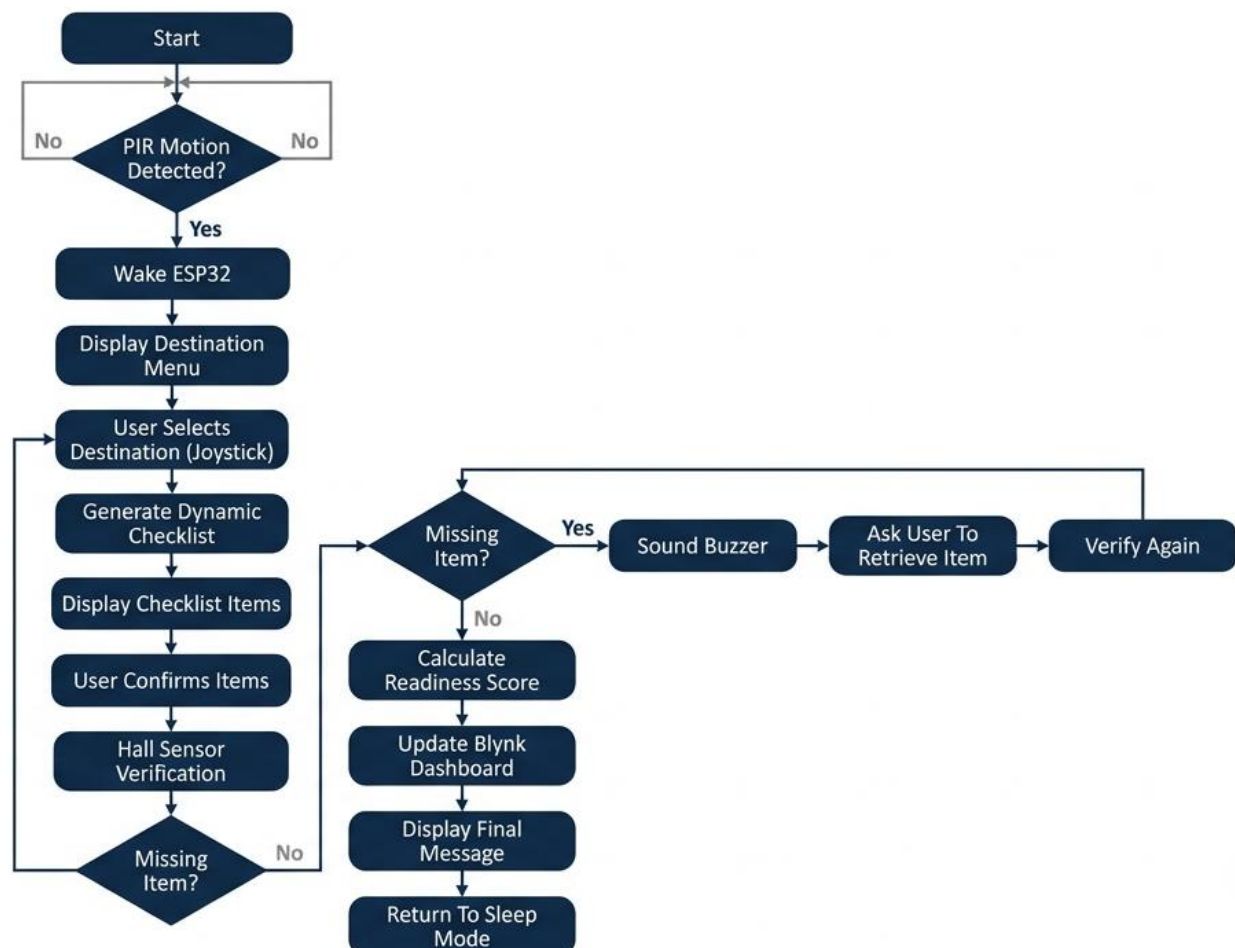


4.6 System Workflow

This section illustrates the complete operational workflow of the Ready2Go system. The following flowcharts summarize the interaction between the user, the embedded controller, the hardware verification subsystem, and the Blynk IoT platform. Instead of describing every software function individually, these diagrams provide a high-level understanding of how the system behaves during normal operation and during custom destination configuration.

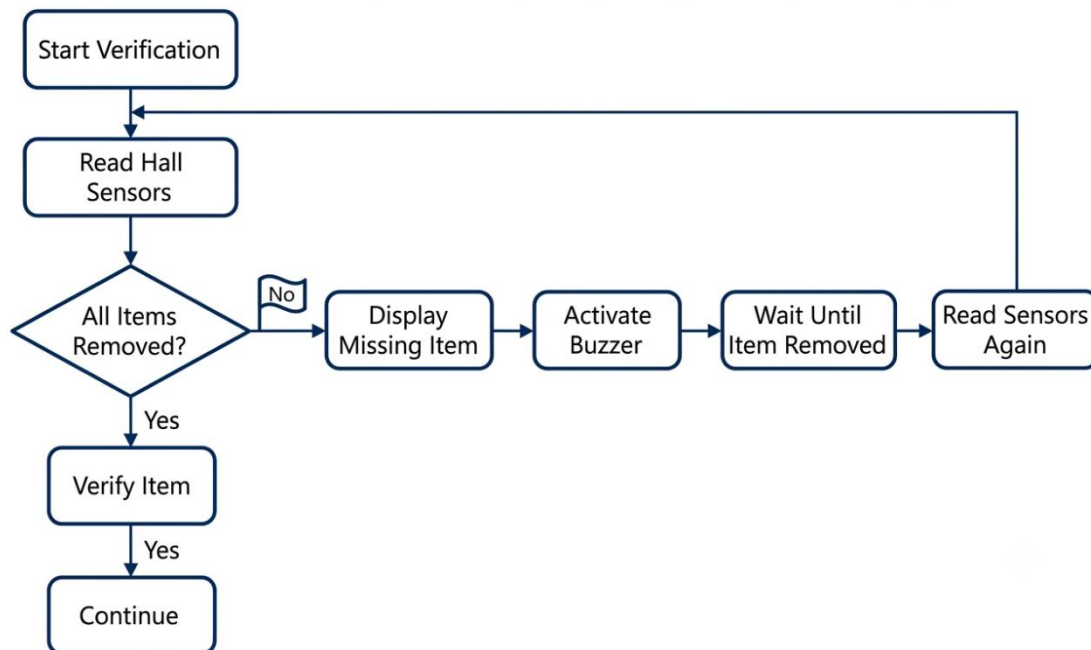
4.6.1 Overall Flowchart

The overall system workflow begins when the PIR sensor detects user presence, activating the ESP32 and waking the system from sleep mode. The user selects a destination using the joystick, after which the firmware dynamically generates the appropriate checklist. The user confirms each required item before the system performs physical verification using Hall-effect sensors. Finally, the readiness score is calculated, the Blynk dashboard is updated, and the system returns to sleep mode waiting for the next user.



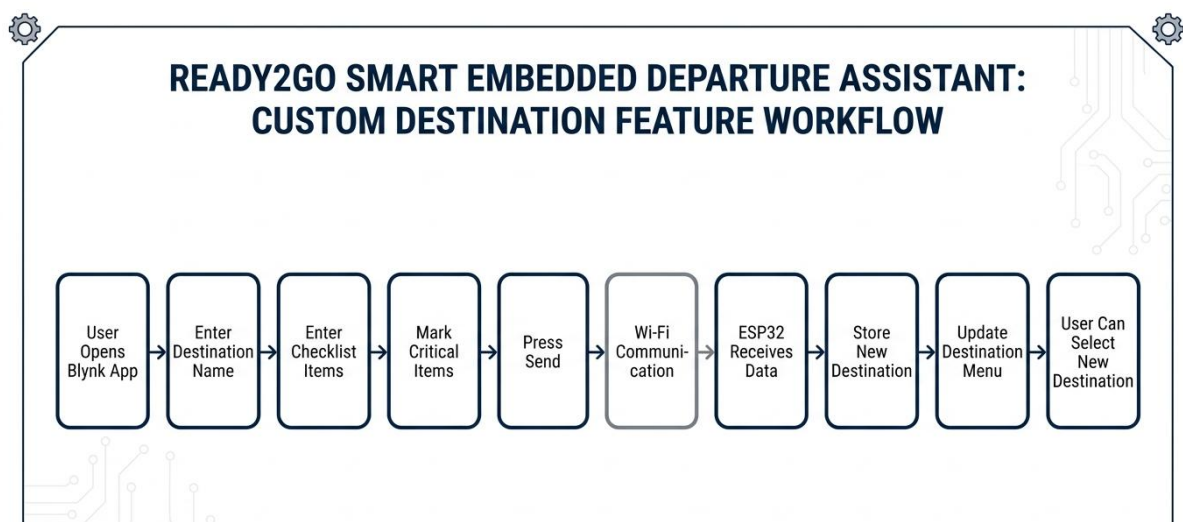
4.6.2 Hardware Verification Flow

Ready2Go introduces a second verification layer after the checklist has been completed. Instead of relying solely on user confirmation, Hall-effect sensors verify whether critical belongings have been removed from their designated locations. If any monitored object remains on the shelf, the system repeatedly activates the buzzer and requests the user to retrieve the missing item before allowing the session to continue.



4.6.3 Custom Destination Workflow

Unlike conventional embedded reminder systems, Ready2Go allows users to create completely new destinations without modifying the firmware. Through the Blynk mobile application, the user specifies a destination name together with its checklist items. ESP32 receives this information over Wi-Fi, stores it in memory, and automatically treats it as a new selectable destination during the next session.



5. Implementation

5.1 Prototype Development

The Ready2Go prototype was developed through an incremental implementation approach, where each hardware module was individually tested before being integrated into the complete embedded system. The development process started with the ESP32 microcontroller and LCD interface to establish the user interaction layer.

After validating the display functionality, the joystick module was integrated to enable destination selection and menu navigation.

The PIR motion sensor was then added to implement automatic system activation upon detecting user presence. Next, the checklist management logic was implemented, followed by the integration of Hall-effect sensors for physical verification of critical belongings. Finally, the Blynk IoT platform was connected to provide wireless monitoring and remote customization features.

This modular development strategy simplified debugging, reduced integration complexity, and ensured that every subsystem operated correctly before forming the complete Ready2Go prototype.



5.2 Hardware Assembly

The hardware prototype was assembled on a solderless breadboard to facilitate rapid development, testing, and future modifications. The ESP32 serves as the central processing unit and interfaces with all peripheral devices.

The LCD module provides real-time feedback to the user, while the joystick and push buttons enable menu navigation and checklist interaction. The PIR sensor is positioned near the entrance to detect user movement and automatically activate the system. Three Hall-effect sensors are installed on the storage locations of essential belongings, where small neodymium magnets are attached to the corresponding objects. This arrangement allows the system to physically verify whether critical items have been removed before departure. An active buzzer provides audible alerts whenever important belongings remain in their designated locations.

The modular wiring layout simplified troubleshooting and ensured reliable communication among all hardware components.

5.3 Software Implementation

The firmware was developed using the Arduino framework for the ESP32 platform. The software follows a modular architecture in which every functional unit is implemented as an independent software module.

The system operates as a finite state machine that transitions through several operating states, including Sleep Mode, Destination Selection, Checklist Execution, Hardware Verification, and Session Completion. This architecture simplifies program flow while improving maintainability and scalability.

Several software modules cooperate during execution, including the motion detection module, destination management module, dynamic checklist generator, decision engine, Hall sensor verification module, readiness score calculator, and Blynk communication interface. The implementation also employs reusable helper functions to reduce code duplication and improve readability.

Overall, the firmware consists of approximately several hundred lines of structured Embedded C++ code designed specifically for real-time embedded operation.



```
Ready2Go.ino
115 void updateStatus(String status) {
116     Blynk.virtualWrite(V0, status);
117 }
118
119 void updateDestination(String dest) {
120     Blynk.virtualWrite(V1, dest);
121     Serial.println("[INFO] Destination = " + dest);
122 }
123
124 void updateChecklistItem(String item) {
125     Blynk.virtualWrite(V2, item);
126     if (item != "Done" && item != "None") {
127         Serial.println("[CHECK] " + item);
128     }
129 }
130
131 void applyPenalty(int penalty) {
132     readinessScore -= penalty;
133     if (readinessScore < SCORE_MIN) readinessScore = SCORE_MIN;
134     Blynk.virtualWrite(V3, readinessScore);
135 }
```

5.4 Blynk Mobile Application

The Ready2Go mobile interface was developed using the Blynk IoT platform to provide real-time interaction with the embedded system. The dashboard communicates with the ESP32 over Wi-Fi and continuously synchronizes the system status during operation.

The application displays essential information such as the current destination, checklist progress, readiness score, missing items, and system status. In addition, users receive real-time assistant messages that describe the current operating stage of the embedded device.

One of the most important features implemented in the application is the **Custom Destination** module. Users can remotely create a completely new destination, define a personalized checklist, and specify critical items directly from the mobile interface. These parameters are transmitted to ESP32 and immediately become available without requiring firmware modifications or reprogramming.

This functionality significantly increases the flexibility and usability of the Ready2Go system.

5.5 Testing Scenarios

To evaluate the functionality of the proposed system, several practical test scenarios were conducted under normal operating conditions. Each subsystem was verified individually before testing the complete workflow.

Test Scenario	Expected Behavior	Result
PIR motion detection	System wakes automatically	Passed
Destination selection	Correct checklist is generated	Passed
Joystick navigation	Menu navigation operates correctly	Passed
Non-critical item skipped	Penalty applied and checklist continues	Passed
Critical item skipped	User warned and item cannot be ignored	Passed
Hall sensor verification	Forgotten belongings detected	Passed
Removing forgotten item	Verification succeeds immediately	Passed
Readiness score calculation	Score updated correctly	Passed
Blynk synchronization	Dashboard updated in real time	Passed
Custom destination	New destination generated successfully	Passed

The experimental evaluation demonstrated that the system successfully performed all intended functions without requiring firmware modifications during operation. Both the embedded hardware and the IoT platform operated synchronously, providing reliable real-time interaction throughout the departure process.

6. Results and Discussion

This chapter presents the experimental evaluation of the Ready2Go prototype under various operating scenarios. The system was tested to verify the functionality of each hardware and software module, evaluate the interaction between subsystems, and assess the effectiveness of the proposed embedded architecture. The obtained results demonstrate the reliability of the prototype in performing destination selection, checklist generation, physical verification, and IoT synchronization while maintaining a smooth and responsive user experience.

6.1 Experimental Results

The prototype was evaluated through multiple practical test scenarios that represent the expected real-world usage of the system. Each subsystem was verified individually before evaluating the complete workflow. The results confirmed that all major system functions operated successfully throughout the testing process.

Test Scenario	Expected Result	Outcome
PIR Motion Detection	Automatically wake the system when user approaches	Passed
Destination Selection	Display the selected destination correctly	Passed
Dynamic Checklist Generation	Generate checklist according to destination	Passed
Critical Item Protection	Prevent users from skipping critical belongings	Passed
Hall Sensor Verification	Detect forgotten physical belongings	Passed
Hardware Re-validation	Stop alarm immediately after removing forgotten item	Passed
Readiness Score Calculation	Calculate score correctly after session completion	Passed
Blynk Dashboard Synchronization	Update dashboard in real time	Passed
Custom Destination Creation	Receive new destination without firmware update	Passed

The experimental evaluation showed that the prototype successfully completed all planned operating scenarios without requiring any firmware modifications during runtime. The integration between the ESP32 firmware, hardware sensors, LCD interface, and Blynk IoT platform remained stable throughout the evaluation process.

6.2 Performance Discussion

The experimental evaluation demonstrated that the proposed system operates reliably under normal usage conditions. The finite state machine architecture ensured smooth transitions between system states while preventing unexpected behavior during user interaction. The destination selection process was responsive, and the dynamic checklist generation mechanism successfully adapted to different destinations without requiring any code modifications.

The Hall-effect sensor verification stage significantly improved the reliability of the system by validating the physical presence of essential belongings instead of relying solely on user confirmation. This additional verification layer successfully detected forgotten critical items and continuously alerted the user until the missing item was removed from its designated location.

Furthermore, the integration with the Blynk IoT platform enabled real-time synchronization between the embedded device and the mobile dashboard. Users were able to monitor the system status, checklist progress, readiness score, and session information while also creating custom destinations remotely without reprogramming the ESP32.

Overall, the prototype achieved stable operation throughout all testing scenarios while maintaining a simple, low-cost, and responsive embedded implementation.

6.3 Comparison with Existing Solutions

Feature	Reminder App	RFID System	Smart Home System	Ready2Go
Destination-Based Checklist	×	×	Limited	✓
Physical Item Verification	×	✓	×	✓
Critical Item Protection	×	Partial	×	✓
Dynamic Checklist Generation	×	×	Limited	✓
IoT Dashboard	Limited	×	Partial	✓
Custom Destination Creation	×	×	×	✓
Firmware-Free Customization	×	×	×	✓
Readiness Score Evaluation	×	×	×	✓

The comparison highlights the advantages of Ready2Go over existing approaches. Unlike conventional reminder applications, the proposed system combines software interaction with physical object verification, significantly reducing the possibility of forgetting essential belongings. Additionally, the integration of customizable destinations through the Blynk platform allows users to extend the system functionality without modifying the embedded firmware, making the solution both flexible and practical for everyday use.

6.4 Discussion

The results obtained demonstrate that Ready2Go successfully fulfills the objectives established at the beginning of this project. The prototype combines embedded sensing, intelligent decision making, and IoT connectivity within a single integrated platform while maintaining low hardware complexity and cost.

One of the key strengths of the system is the use of Hall-effect sensors to verify the actual removal of important belongings, eliminating the limitations of conventional reminder systems that rely entirely on user input. Moreover, the customizable destination feature increases the adaptability of the system, allowing users to personalize their departure routines without additional programming.

Although the current prototype focuses on a limited number of essential belongings, the modular hardware and software architecture allows additional sensors, destinations, and intelligent features to be incorporated with minimal modifications. This demonstrates the scalability of the proposed design and its potential for future development into a more comprehensive smart assistant.

7. Future Work

Although the proposed Ready2Go system successfully demonstrates the integration of embedded sensing, physical verification, and IoT connectivity, several improvements can further enhance its functionality and usability. The modular architecture of the system allows additional features to be incorporated with minimal modifications to both hardware and software.

One possible enhancement is the integration of artificial intelligence to analyze users' daily routines and automatically recommend checklist items based on previous behavior, departure time, or destination history. Such personalization could significantly improve the user's experience.

Another promising extension is synchronization with digital calendars and weather services. By accessing scheduled events and weather forecasts, the system could automatically generate destination-specific reminders, such as carrying an umbrella on rainy days or bringing presentation materials before meetings.

Future versions may also include voice interaction through speech recognition and voice synthesis, allowing users to communicate with the system without using physical buttons. This would improve accessibility and provide more natural interaction.

The current prototype verifies only a limited number of essential belongings using Hall-effect sensors. Additional sensing technologies, such as RFID, NFC, Bluetooth tags, or computer vision, could be integrated to support a wider range of personal items while increasing verification accuracy.

From the IoT perspective, cloud-based analytics can be introduced to record user sessions, identify frequently forgotten items, and generate long-term statistics that help users improve their departure habits.

Finally, future hardware revisions may include a custom PCB, rechargeable battery operation, and a compact 3D-printed enclosure to transform the prototype into a portable commercial-grade-embedded product.

8. Conclusion

This project presented **Ready2Go**, a smart embedded departure assistant designed to reduce the common problem of forgetting essential belongings before leaving home. By combining embedded sensing, intelligent decision-making, physical object verification, and IoT connectivity, the proposed system provides a practical solution that extends beyond the capabilities of traditional reminder applications.

The developed prototype successfully integrates an ESP32 microcontroller with multiple hardware modules, including a PIR motion sensor, Hall-effect sensors, joystick-based navigation, an LCD interface, and an active buzzer. The implementation of a finite state machine ensured reliable system operation, while the Blynk platform enabled real-time monitoring and remote customization through a mobile application.

Experimental evaluation confirmed the successful operation of all major system functions, including automatic motion detection, destination-based checklist generation, critical item protection, physical verification of belongings, readiness score calculation, and dynamic creation of custom destinations without modifying the embedded firmware.

Overall, Ready2Go demonstrates how embedded systems can be combined with IoT technologies to deliver an intelligent, user-friendly, and extensible smart assistant. The modular architecture, low hardware cost, and scalable software design make the proposed solution suitable for further development into a real-world commercial product and provide a strong foundation for future research in smart embedded assistance systems.

References

- Arduino. (n.d.). *Arduino Language Reference*. Retrieved August 3, 2026, from <https://www.arduino.cc/reference/en/>
- Arduino. (n.d.). *Arduino Documentation*. Retrieved August 3, 2026, from <https://docs.arduino.cc>
- Blynk Inc. (n.d.). *Blynk IoT platform documentation*. Retrieved August 3, 2026, from <https://docs.blynk.io>
- Espressif Systems. (2025). *ESP32 series datasheet (Version 5.2)*. https://documentation.espressif.com/esp32_datasheet_en.html
- Espressif Systems. (n.d.). *ESP-IDF programming guide*. Retrieved August 3, 2026, from <https://docs.espressif.com/projects/esp-idf/>
- NXP Semiconductors. (2021). *UM10204 I²C-bus specification and user manual (Rev. 7.0)*. <https://www.nxp.com/docs/en/user-guide/UM10204.pdf>
- SparkFun Electronics. (n.d.). *Hall-Effect Sensor Hookup Guide*. Retrieved August 3, 2026, from <https://learn.sparkfun.com/tutorials/hall-effect-sensor-hookup-guide>
- Adafruit Industries. (n.d.). *PIR Motion Sensor*. Retrieved August 3, 2026, from <https://learn.adafruit.com/pir-passive-infrared-proximity-motion-sensor>
- Wokwi. (n.d.). *Wokwi online simulator for ESP32 and Arduino*. Retrieved August 3, 2026, from <https://wokwi.com>
- Zanella, A., Bui, N., Castellani, A., Vangelista, L., & Zorzi, M. (2014). Internet of Things for smart cities. *IEEE Internet of Things Journal*, 1(1), 22–32. <https://doi.org/10.1109/JIOT.2014.2306328>

Appendices

- **Appendix A**

ESP32 Pin	Connected Component	Description
GPIO27	PIR Motion Sensor	Motion Detection
GPIO34	Joystick Y-axis	Menu Navigation
GPIO32	Joystick Switch	Destination Selection
GPIO13	YES Button	Confirm Item
GPIO5	NO Button	Skip Item
GPIO23	BACK Button	Return to Menu
GPIO25	Hall Sensor (Keys)	Physical Verification
GPIO26	Hall Sensor (Wallet)	Physical Verification
GPIO14	Hall Sensor (Student ID)	Physical Verification
GPIO12	Active Buzzer	Audio Alert

- **Appendix B**

Component	Quantity	Purpose
ESP32 DevKit V1	1	Main Controller
LCD 16×2 I2C	1	User Interface
PIR Motion Sensor HC-SR501	1	Motion Detection
Analog Joystick	1	Destination Navigation
Push Buttons	3	Yes / No / Back
Hall Effect Sensor A3144	3	Physical Verification
Neodymium Magnets	3	Hall Sensor Detection
Active Buzzer	1	Alerts
Breadboard	1	Prototype
Jumper Wires	Set	Connections
USB Cable	1	Programming & Power

- **Appendix C**

Virtual Pin	Function
V0	System Status
V1	Current Destination
V2	Current Checklist Item
V3	Readiness Score
V4	AI Assistant Messages
V5	Missing Items
V6	Custom Destination Name
V7	Custom Checklist

- **Appendix D**

**GitHub Repository:**

<https://github.com/AhmedSamirNU/Ready2Go>

**YouTube Demonstration Video:**

https://youtu.be/xozcsi4o3Z0?si=FPg_lo4ks2JNHsAm

**Wokwi Simulation:**

<https://wokwi.com/projects/470927609513027585>

**Blynk Template Name:**

Ready2Go

- **Appendix E**

Item	Value
Microcontroller	ESP32 DevKit V1
Supply Voltage	5V USB
Programming Language	Arduino C++
IDE	Arduino IDE 2.x
IoT Platform	Blynk IoT
Communication	Wi-Fi 2.4 GHz
Development Simulator	Wokwi